

MULTI-OBJECT TRACKING RADAR SYSTEMS TECHNICAL REQUIREMENTS DOCUMENT

1. Background

The Eglin Gulf Test and Training Range (EGTTR) currently utilizes legacy single-object tracking radars to provide safety level time/space/position information (TSPI) to relevant USAF test stakeholders. The 96th Test Wing is interested in understanding current Commercial-Off-The-Shelf (COTS) and Government-Off-The-Shelf (GOTS) solutions for accurate TSPI systems to support multiple simultaneous systems under test; both cooperative and non-cooperative. The Multiple Object Tracking Radars (MOTRs) will be used to provide accurate TSPI tracks for all airborne test platforms, test items, and targets utilizing the Eglin Test and Training Complex (ETTC) test space for airborne missions. Modern radar systems are complex, custom OEM manufactured items. All engineering and technical information for these systems (drawings, specifications, performance data, software coding languages and software programming, interface communication protocols) is owned by and proprietary to the OEM and not owned by the government. Proof of direct applied knowledge of or ability to obtain ready access to OEM proprietary engineering and technical information is required.

2. Technical Requirements

2.1 Ability to track multiple high-speed targets (20 simultaneous tracks without accuracy degradation).

2.2 Radar shall track targets moving at radial velocities up to 6600 knots (Mach 10) and angular rates up to 30 degrees per second.

2.3 Radar shall be capable of performing both skin tracking and c-band tracking.

2.4 Radar shall autotrack all discernable objects within the main beam.

2.5 Radar shall automatically detect and track separating objects originating from a primary object already being tracked.

2.6 Radar shall track targets with an angular tracking acceleration up to 45 degrees per second squared.

2.7 Support diverse, spatial, and temporal situation awareness for range safety.

2.8 Radar shall be capable of detecting, acquiring, and skin tracking a non-cooperative target of 1 square meter with 0 dBsm radar cross section (RCS) at a range of 130 nautical miles with a signal-to-noise ratio (SNR) of ≥ 12 dB.

2.9 Radar shall be capable of tracking two objects in three axes to an accuracy of 10 meters RMS in range and 0.1 degrees RMS in azimuth and elevation angles at SNR ≥ 20 dB.

2.10 Beacon Track: > 500nmiles (Beacon compliant with RCC Standard 262-14 [C (G) – Band & X (I)])

2.11 Frequency: 5.4 – 5.9 GHz

2.12 Mobility: Trailer Mounted; Suitable for long-distance highway transportation

2.13 Remote Control: Via digital network

2.14 Operating System: Windows 11 or equivalent/better; upgradeable

2.15 Range Accuracy: $\leq 15'$, 1 sigma

2.16 Angle Accuracy: ≤ 0.1 milliradian, 1 sigma

2.17 Azimuth Coverage: 360 deg continuous

2.18 Elevation: -5 to 185 deg

2.19 Ability to withstand harsh mechanical, electrical, and weather environments.

2.20 Slaving: Shall provide either tracked target locations to any other system or accept target data from any other system.

2.21 Calibration: Range and Az/El calibration per IRIG standard; method to perform multi-track calibration.